

Recorded salutation from wearable epoch data

Cohort design, feature representations, random-forest experiments, and a proposed analysis of informative missingness

Central finding. Wearable epoch data contain modest information about recorded salutation. Most usable signal is tied to recording pattern, device context, and broad heart-rate summaries. Regularization helps. Feature selection and the tested hand-built transformations do not provide a reliable general improvement.

Evidence base: frozen September 2026 artifacts in `artifacts/`, `artifacts_v2/`, and `artifacts_sq/`.

Outcome: recorded salutation code 10 versus 20.

Status: all model results are completed; the missingness analysis is a proposal and has not been run.

Executive summary

The experiments answer a prediction question: how well can summaries of wearable recordings separate two recorded salutation codes among participants represented in this export? They do not identify biological sex, gender identity, or causal physiology.

0.713

Pooled v1 frozen-test AUROC; balanced accuracy 0.590.

0.55–0.60

Demographic-only frozen-test AUROC across the three strict source cohorts.

≤ 0.0034

Largest mean primary feature-engineering gain; its simultaneous interval includes zero.

54.2% vs 33.1%

Share recorded as salutation 10 below the coverage gate versus among gate passers.

Best-supported reading: the models use a mixture of recording behavior, device or vendor context, availability of processed channels, and heart-rate summaries. The current experiments cannot isolate a physiological component.

1. The pooled baseline showed above-chance but modest discrimination and strongly asymmetric class recall.
2. Within source-specific cohorts, recording and heart-rate summaries clearly outperformed age/BMI alone. Which time representation worked best differed by source.
3. Cross-validated regularization—especially a larger minimum leaf size—improved the random forest. A single fixed regularized configuration was nearly as good as cell-specific tuning.
4. Across repeated splits, feature hygiene greatly reduced width without a detected mean AUROC change. Correlation pruning, top-k selection, and the tested engineered terms were not consistently useful.
5. The next useful diagnostic is to model the observation process directly, including participants below the coverage gate. This should be described as an informative-missingness study, not as proof that the data are MNAR.

1. Study question and data

The outcome was the recorded salutation field: code 10 was mapped to class 0 and code 20 to class 1. Code 30 and missing labels were excluded from fitted models. Core epoch channels were 3000 (heart rate), 3001 (resting heart rate), and 3002 (hourly resting heart rate). Measurements outside 25–230 bpm were excluded. Participant identity, not rows or days, defined every train/validation/test split.

For the pooled cohort, an adequate channel-day required at least 8 observed clock hours and either 8 hours of interval coverage or 60 valid measurements. A participant passed when at least one channel had 14 adequate days.

Scope: results are conditional on participants with observed labels and on the export, source mix, coverage rules, and analysis choices. Vendor-derived channels may themselves use proprietary processing or user-entered information. This is a classification study, not a biological or causal study.

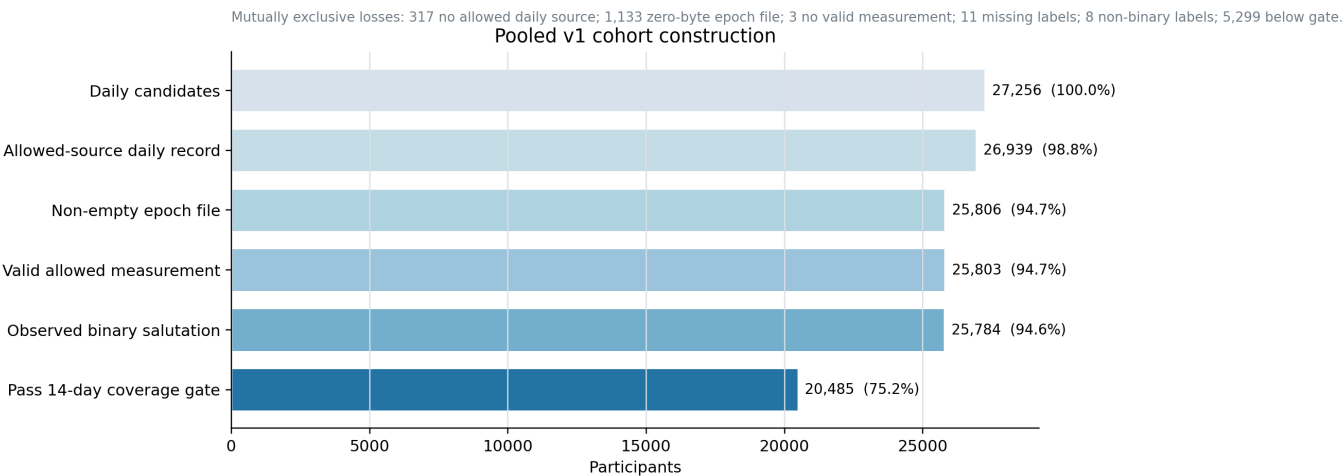
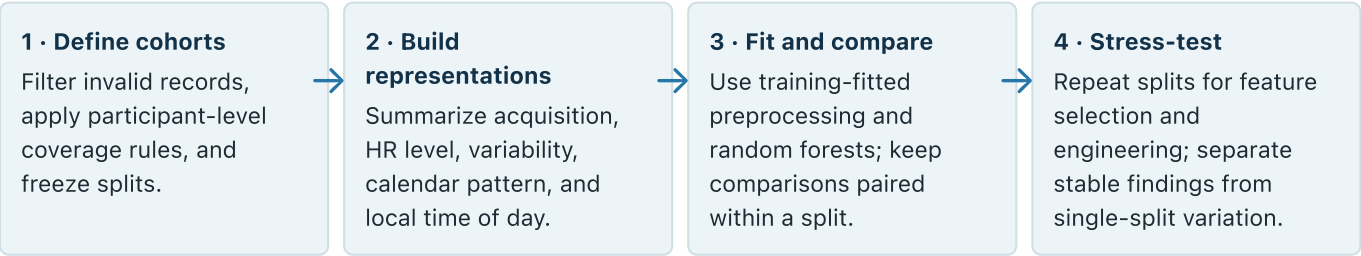


Figure 1. Pooled v1 cohort construction. The coverage gate removed 5,299 otherwise labelled participants, making the fitted cohort a selected population.

Experiment outline

Stage	Population and split	Question	Status
Pooled v1	20,485 gate-passing participants; 80/10/10	Can epoch summaries predict recorded salutation?	Complete
Pooled v2	20,423 after extra source exclusions; 80/10/10	Do demographics and source membership change the pooled result?	Complete
Strict per-source	Sources 3/6/7; sole-source 91-day gate; 80/10/10	Which time representation works within each source?	Complete
Tuning	Same frozen per-source test sets; CV on train+validation	Can regularization repair the default RF?	Complete
Feature selection	30 repeated 70/15/15 splits per source	Do hygiene, deduplication, or top-k selection help?	Complete
Feature engineering	Same 30 splits and RF seeds	Do simple interactions, ratios, or drift terms help?	Complete
Missingness study	Gated and ungated pooled cohorts	How much signal is in the observation process itself?	Proposed, not run



2. Feature representations

The pooled experiments used one broad participant-level table. The side quest separated the same general information into blocks so that time scope and representation could be compared within each source.

Representation	Raw inputs	Plain-language description
Demographics (`demo`)	2	Age group and BMI group. Reference model; included once in every side-quest design.
Full recording (`rec`)	80	Demographics plus 78 acquisition, coverage, HR level, variability, weekday/weekend, and time-of-day summaries over the available record.
Selected window (`win`)	80	The same summary families inside the selected 91-day sustained-recording window.
Rolling full (`roll_rec`)	122	Demographics plus 7- and 30-day rolling summaries across the recording span.
Rolling window (`roll_win`)	122	Rolling endpoints inside the selected window, with up to 29 days of pre-window lookback.
All blocks (`all`)	398	Demographics and all four epoch blocks combined.

Numeric preprocessing was learned on training participants only. The original and default side-quest models used median imputation and per-column missingness indicators, followed by zero-variance removal. Categorical variables used most-frequent imputation and one-hot encoding. No scaling was required for random forests.

Important distinction: acquisition variables such as observed days, coverage, event counts, span, and channel availability describe how data were recorded. HR means and variability describe recorded values. They are correlated but are not the same kind of information.

3. Pooled baseline experiments

Experiment	N	Main change	Validation AUROC	Test AUROC	Test balanced accuracy	Bootstrap SD
Pooled v1	20,485	89 epoch/recording predictors; sources 38/46/48 excluded	0.668	0.713	0.590	0.016
Pooled v2	20,423	Adds age/BMI and source indicators; also excludes sources 2/4/19	0.706	0.683	0.581	0.009

Both models separated the classes better than chance, but default 0.5-threshold predictions mostly favored the more common class 1. In v1, class-0 recall was 0.255 and class-1 recall was 0.924. The corresponding v2 recalls were 0.246 and 0.915.

The v1 and v2 test AUROCs should not be subtracted as if they were a paired treatment comparison. V2 changed the cohort, split, source exclusions, and predictors simultaneously. Its lower test point does, however, show that simply adding age/BMI and source-membership indicators did not create an obvious improvement.

4. Strict per-source side quest

The side quest required a sustained 91-day window within a single source. Within one channel, a strict day required at least 12 hours of the frozen coverage quantity; a passing window required at least 65 strict days and at least 11 of 13 weeks with five strict days. Eligibility was assessed separately for each core channel; days were never pooled across channels. Sources 3, 6, and 7 had at least 200 participants in each label class and were modelled. Sources 9 and 13 remained audit-only.

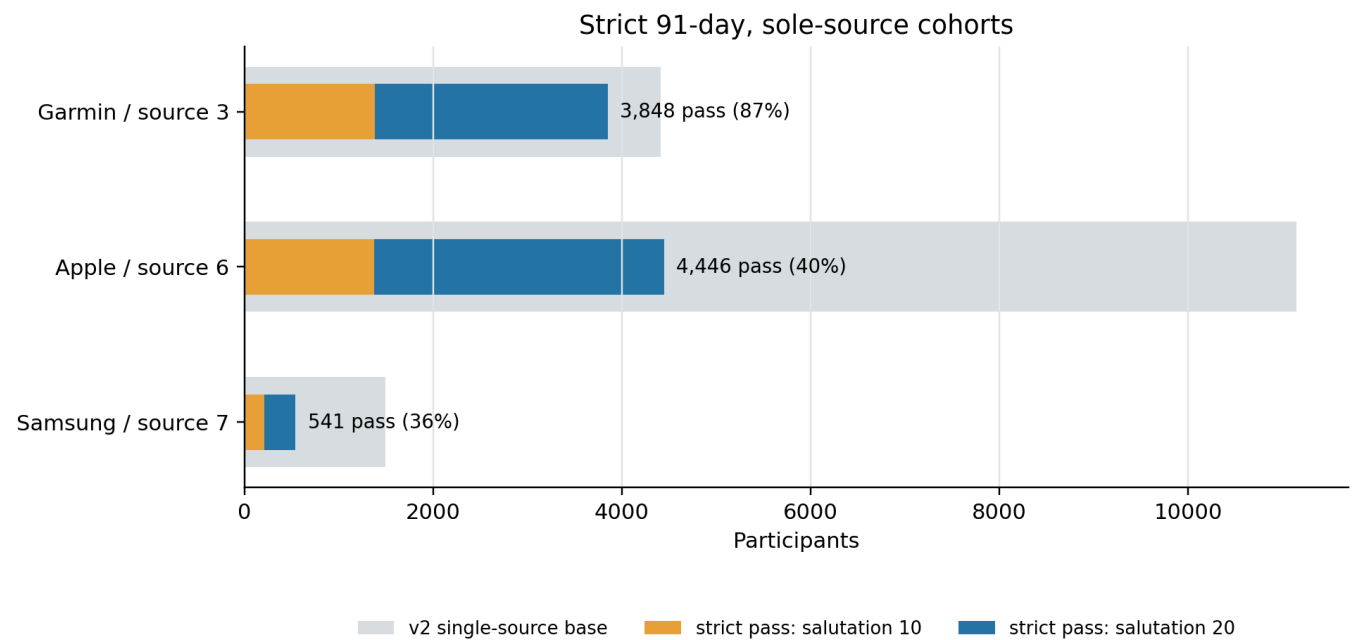


Figure 2. Strict source-specific cohorts. Gray bars show the v2 single-source base; colored bars show strict-pass participants by label. Source 6 retains a much smaller fraction; 4,367 of its 4,446 passers qualify through channel 3002.

Source cohort	Strict-pass N	Salutation 10	Salutation 20	Frozen test N	Retention from sole-source base
Garmin / source 3	3,848	1,384	2,464	385	87.3%
Apple / source 6	4,446	1,373	3,073	444	39.9%
Samsung / source 7	541	209	332	54	36.1%

Frozen-split results

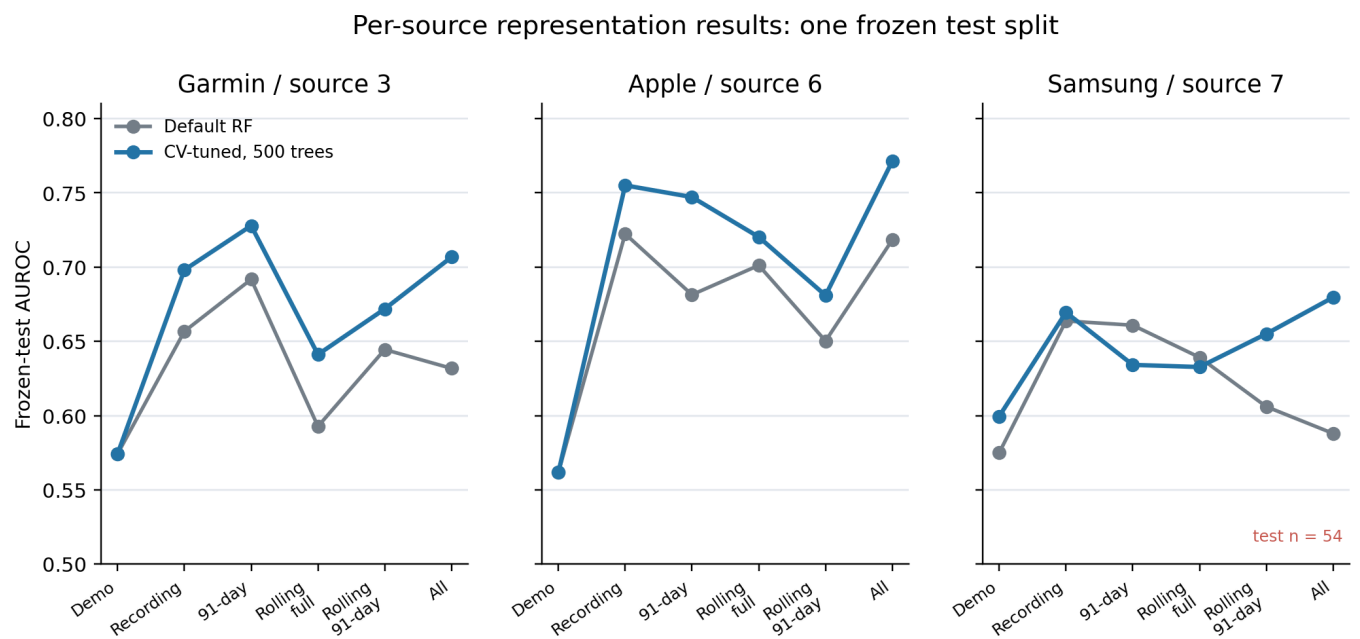


Figure 3. Default and tuned random forests on the same frozen source-specific test folds. These are descriptive point estimates. Source 7 has only 54 test participants, so its apparent ordering is especially unstable.

Source cohort	Default demo	Default recording	Default 91-day	Default all	Largest tuned test point
Garmin / source 3	0.574	0.657	0.692	0.632	win (0.728)
Apple / source 6	0.562	0.722	0.681	0.718	all (0.771)
Samsung / source 7	0.575	0.664	0.661	0.588	all (0.680)

What these comparisons show

- **Demographics alone were weak.** Their frozen-test AUROC was about 0.56–0.60.
- **Recording and epoch summaries carried most of the prediction.** The recording representation improved substantially over demographics in sources 3 and 6, and less precisely in source 7.
- **The useful time scope was source-specific.** The 91-day block worked well for source 3; the full recording and combined blocks were stronger for source 6 after regularization.
- **More columns were not automatically better.** Under default RF settings, the all-block design could be diluted by wide, redundant inputs.

5. Tuning: regularization mattered

The tuning extension searched 48 random-forest configurations using five-fold cross-validation on each train-plus-validation pool. The frozen test fold was not used for selection. Final models used 500 trees.

10

The larger minimum leaf size was the clearest and most consistent useful setting.

≤ 0.006

CV AUROC gap between the fixed G1 configuration and each searched cell optimum.

A practical shared setting emerged: `max_features=0.4`, `min_samples_leaf=10`, unrestricted depth, and `class_weight=balanced_subsample`. The default sklearn configuration ranked near the bottom of the complete tuning surfaces. The main improvement therefore came from regularization, not merely from using 500 rather than 100 trees.

The most defensible performance gain is the approximately 0.015–0.020 increase in cross-validated AUROC over the default settings. Frozen-test gains are supportive but remain a second look at one split. The later repeated-split phases used the shared regularized configuration with 100 trees to control compute.

6. Repeated-split feature selection and engineering

Both phases used 30 stratified 70/15/15 participant splits per source and the same RF seeds for paired arms. Their estimand is the average effect over the prescribed split/model randomization, conditional on the observed strict-coverage cohort. Reusing these participants and partitions does not provide independent validation.

Feature selection

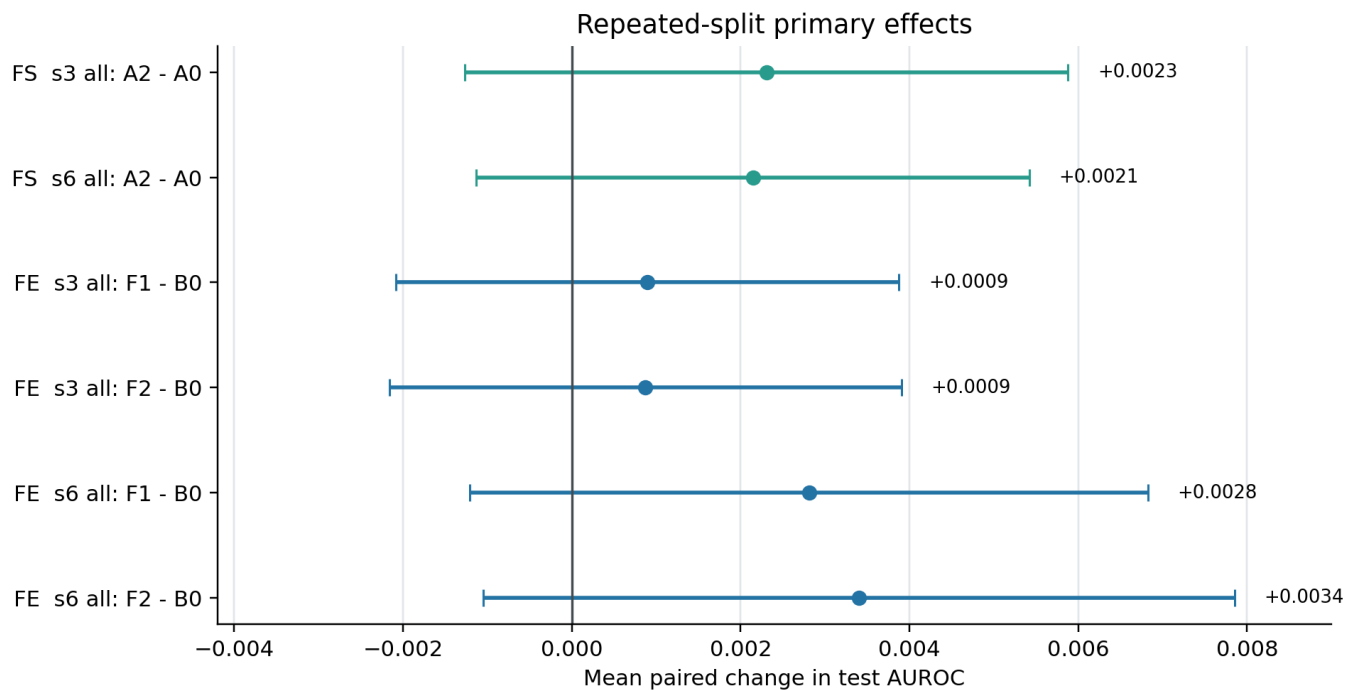
- **A0:** standard preprocessing.
- **A1:** remove missingness-indicator columns and near-zero-variance terms.
- **A2:** A1 plus greedy correlation pruning at $|\rho| > 0.95$.
- **A3:** top 50 or 100 terms selected by validation-fold permutation importance.

A1 produced no detected mean AUROC change in any source-by-representation cell while markedly reducing the widest matrices. This supports it as a computational hygiene step, but it is not a formal equivalence result and does not isolate indicator removal from near-zero-variance removal. A2 and A3 produced small, source-dependent effects; top-k sets were unstable across splits.

Feature engineering

The tested families were: products of weekday mean HR and time-of-day mean HR (E1, 24 terms), channel ratios (E2, 12 terms), and 30-day minus 7-day rolling changes (E3, 18 terms). These are deterministic re-expressions of existing measurements; they may make a tree split easier but do not add a new data source.

No primary comparison showed a detectable general AUROC improvement. Some secondary cells moved in either direction, but the effects did not transport consistently across sources or related representations.



FS: 97.5% intervals for its 2-comparison family. FE: 98.75% intervals for its 4-comparison family. All intervals include zero.

Figure 4. Corrected simultaneous intervals for the declared repeated-split primary families. The feature-selection family has two comparisons; the feature-engineering family has four. Every interval crosses zero.

Primary comparison	Mean delta	Simultaneous interval	Per-comparison level
Feature selection, s3 all, A2 - A0	+0.0023	[-0.0013, +0.0059]	97.5%
Feature selection, s6 all, A2 - A0	+0.0021	[-0.0011, +0.0054]	97.5%
Feature engineering, s3 all, F1 - B0	+0.0009	[-0.0021, +0.0039]	98.75%
Feature engineering, s3 all, F2 - B0	+0.0009	[-0.0022, +0.0039]	98.75%
Feature engineering, s6 all, F1 - B0	+0.0028	[-0.0012, +0.0068]	98.75%
Feature engineering, s6 all, F2 - B0	+0.0034	[-0.0010, +0.0078]	98.75%

Decision: keep the simpler hygiene preprocessing for efficiency. Do not adopt correlation pruning, top-k selection, or these engineered families as general defaults. This is a decision about the tested methods, not proof that every possible representation or model would fail.

7. Proposed next study: recording missingness as signal

Status: not run. The existing coverage rule discards participants with sparse recordings, yet gate failure is strongly associated with the recorded outcome. The observation process may therefore contain useful predictive information. The right question is not "can MNAR be proved?" but "how much discrimination is carried by recording availability and pattern?"

Coverage and recording availability are associated with the recorded label

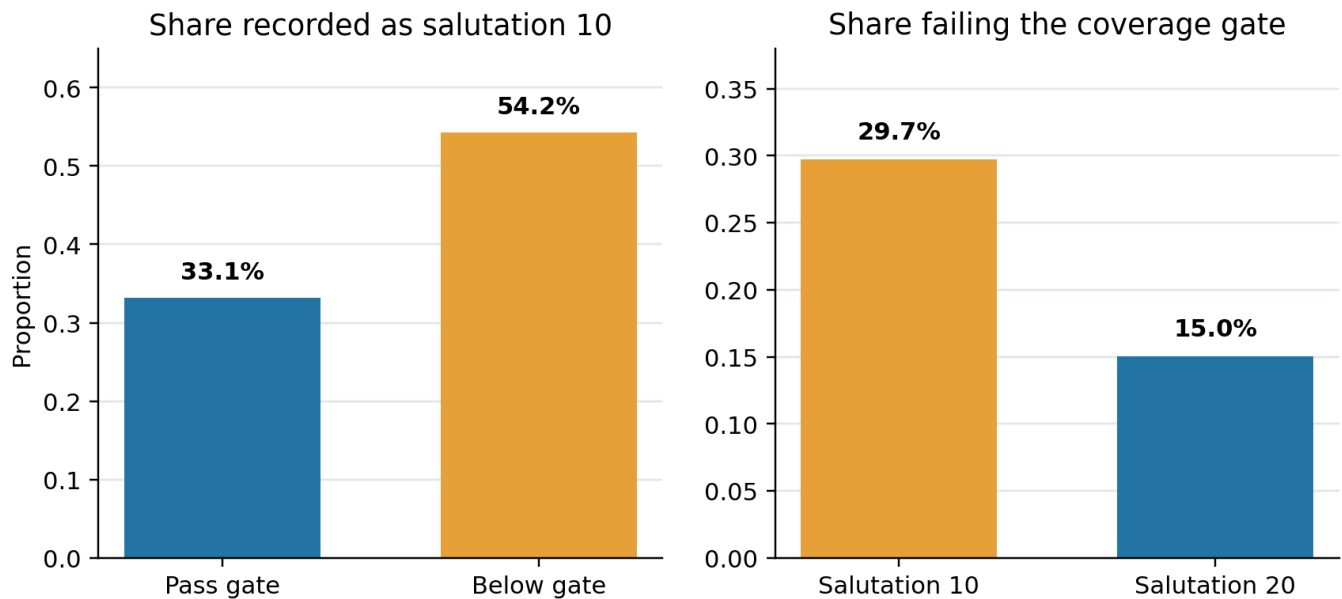


Figure 5. Observed association motivating the missingness study. This is descriptive and does not identify why recording differs.

Two target populations

Code	Participants	Definition	Interpretation
G: gated	20,485	Existing v1 gate passers; reuse the frozen split	Performance among sustained recorders. The old test set has already informed later choices, so this is exploratory.
U: ungated	25,784	Gate passers plus 5,299 labelled participants below the gate; freeze a new split	Performance in the broader measured population, with gate status treated as a predictor rather than an exclusion.

Simple recommended implementation

1. **Build a recording-process block.** Use observed and adequate days, event counts, span, duty cycle, longest gaps, hours covered, channel presence, time-of-day event fractions, and device/source fields.
2. **Keep a value block.** Use HR level and variability summaries. With current frozen artifacts, time-of-day means must use all valid events or be omitted; adequate-day-only diurnal means cannot be recovered because the hourly table has no date dimension.
3. **Fit small, nested comparisons.** Include source-only and gate-only anchors, recording-process only, HR values only, and their combination. This separates “mostly device/gate” from richer recording-pattern signal.
4. **Use two value encodings.** A train-median, no-indicator version largely suppresses missingness information; a native-NaN version allows the RF to use missingness implicitly. Label the latter honestly as “values plus implicit missingness.”
5. **Keep evaluation participant-level and paired.** Fit preprocessing on training participants only. Reuse common resamples across variants. If only ten bootstrap draws are retained for direct v1 compatibility, treat them as a rough spread, not an interval.

Candidate model	Inputs	What it answers
Source/gate anchors	Source and multisource; gate status in U	How much prediction comes from coarse device or selection information alone?
R	Detailed recording-process features, reported with and without source	Do coverage, gaps, timing, and channel presence add information beyond source?
V-imputed	HR values; training-median fill; no missing bits	A practical value-focused comparator with missingness mostly suppressed.
V-native	HR values with NaNs passed to the RF	Values plus implicit per-feature missingness.
R + V-native	Both blocks	Best missingness-aware prediction using the existing data.

Interpretation limit: this design can show that missingness or recording patterns are predictive. It cannot establish an MNAR mechanism, recover the 11 missing outcomes, or show that the signal is physiological. A value can be missing because of wear, device capability, software, study engagement, or data transfer.

Other possible implementations

The simplest alternative is the existing median-imputation pipeline with explicit missingness indicators. It is easy to audit but can become wide and the prior strict-cohort indicators were mostly unused. A more complex two-model approach—one model for recording pattern and one for measured values, with their scores combined on validation data—could make block contributions clearer, but it is not necessary for the first pass.

8. What can and cannot be concluded

Supported conclusions

- Epoch-derived participant summaries predict recorded salutation above chance in these selected cohorts.
- Age and BMI groups alone explain little of the source-specific discrimination.
- Recording context and broad HR summaries are the strongest available feature families.
- RF regularization is more useful than cell-specific feature selection or the tested hand-built terms.
- The coverage gate changes the target population and removes an outcome-associated part of the observation process.

Not supported

- A biological-sex or gender interpretation of the classifier.
- A causal physiological effect or a vendor ranking.
- Transport to new participants, devices, software versions, or recording policies without external evaluation.
- Claims that feature selection or engineering is exactly equivalent to baseline; the studies establish a resolution limit, not equality.
- Claims that all model-side improvements are exhausted.

Most valuable next evidence: a pre-frozen ungated missingness analysis followed by evaluation on genuinely new participants or an external export.

9. Verification and provenance

- The pooled v1 and v2 verification artifacts report that all declared checks passed.
- The strict per-source phase reports 135/135 checks passed, covering eligibility, windows, splits, features, predictions, and bootstrap reconstruction.
- The feature-selection and feature-engineering result tables each contain 2,700 complete rows with 30 repeats per recorded group and no duplicate keys or non-finite metrics.
- Feature-engineering arm B0 reproduces feature-selection arm A1 to a maximum AUROC difference of $5.6e-17$.
- This meta-report recomputed its tables and figures from saved JSON/CSV/Parquet artifacts. It fitted no models.

Primary artifact index

Topic	Artifact
Pooled experiment	REPORT.md; artifacts/metrics.json; artifacts_v2/metrics.json
Strict source cohorts	artifacts_sq/source_cohorts.csv; artifacts_sq/sidequest_report.md
Tuning	artifacts_sq/tuned_500_comparison.md; source_*/tuned_500/
Feature selection	artifacts_sq/fsplit/fsplit_metrics.csv; feat_sel_report.md
Feature engineering	artifacts_sq/feng/feng_metrics.csv; feat_eng_report.md
Missingness proposal	MNAR_PLAN.md

Reporting note: this synthesis uses corrected simultaneous confidence levels for the repeated-split primary families and describes E1 using the actual source feature semantics (time-of-day mean HR, not event share). Secondary repeated-split results remain exploratory and are not used to declare winners.